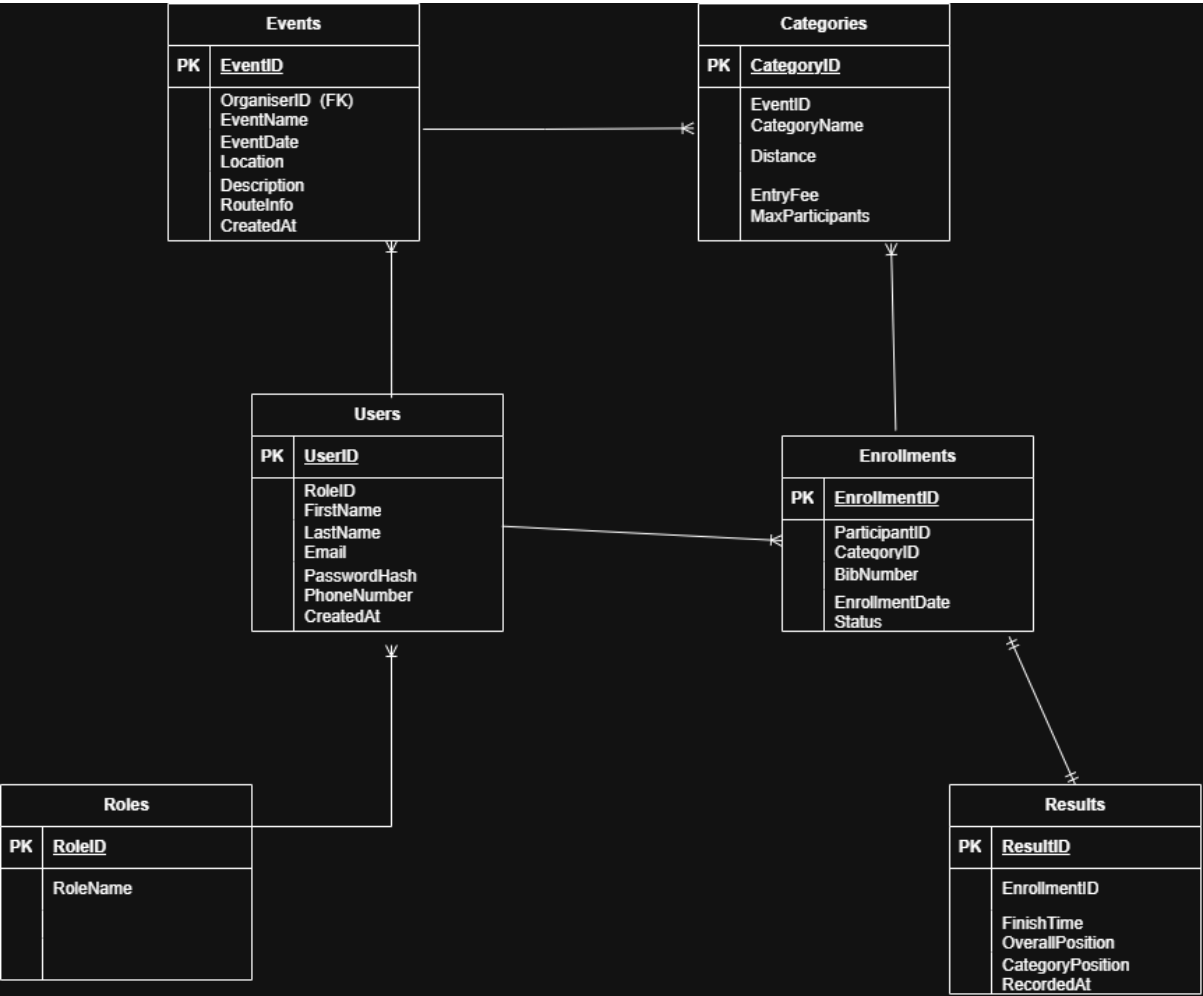


Project Overview

This document contains a system plan of a full-stack web-based event management system that is designed specifically for the South African road running, cycling, and walking community.

This platform will allow event organisers to create and manage events, categories, participants, and results, while participants can browse upcoming events, enrol in events and check their performance history and prepare for race day.

Race Day ERD



API Endpoint Plan

#	HTTP Method	Route	Description	Role Required	Request Body	Expected Response
Authentication						
1	POST	<code>`/api/auth/register`</code>	Registers a new user as either a Participant or an Organiser	Public	<code>`{ firstName, lastName, email, password, phoneNumber, role }`</code>	<code>`201 Created` - { userId, email, role, token }`</code>
2	POST	<code>`/api/auth/login`</code>	Authenticates a user and issues a JWT	Public	<code>`{ email, password }`</code>	<code>`200 OK` - { userId, role, token, expiresAt }`</code>
User Profile						
3	GET	<code>`/api/users/me`</code>	Retrieves the logged-in user's own profile	Participant/Organiser	-	<code>`200 OK` - { userId, firstName, lastName, email, phoneNumber, role, createdAt }`</code>
4	PUT	<code>`/api/users/me`</code>	Updates the logged-in user's own profile details	Participant/Organiser	<code>`{ firstName, lastName, phoneNumber }`</code>	<code>`200 OK` - updated user object`</code>
5	PUT	<code>`/api/users/me/password`</code>	Changes the logged-in user's password	Participant/Organiser	<code>`{ currentPassword, newPassword }`</code>	<code>`204 No Content`</code>
Events						
6	GET	<code>`/api/events`</code>	Lists all upcoming events (supports optional <code>`?location=`</code> and <code>`?date=`</code> filters)	Public	-	<code>`200 OK` - array of event summaries`</code>
7	GET	<code>`/api/events/{eventId}`</code>	Retrieves full details for a single event, including its categories	Public	-	<code>`200 OK` - event object with nested categories`</code>
8	POST	<code>`/api/events`</code>	Creates a new event	Organiser	<code>`{ eventName, eventDate, location, description, routeInfo }`</code>	<code>`201 Created` - created event object`</code>
9	PUT	<code>`/api/events/{eventId}`</code>	Updates an event owned by the logged-in organiser	Organiser	<code>`{ eventName, eventDate, location, description, routeInfo }`</code>	<code>`200 OK` - updated event object`</code>
10	DELETE	<code>`/api/events/{eventId}`</code>	Deletes an event owned by the logged-in organiser	Organiser	-	<code>`204 No Content`</code>
11	GET	<code>`/api/events/{eventId}/weather`</code>	Returns a live weather forecast for the event's location and date, to help participants prepare for race day	Public	-	<code>`200 OK` - { eventId, forecastDate, tempC, conditions, windSpeed }`</code>
Categories						
12	GET	<code>`/api/events/{eventId}/categories`</code>	Lists all categories for a specific event	Public	-	<code>`200 OK` - array of categories`</code>
13	POST	<code>`/api/events/{eventId}/categories`</code>	Adds a new category to an event owned by the logged-in organiser	Organiser	<code>`{ categoryName, distanceKm, entryFee, maxParticipants }`</code>	<code>`201 Created` - created category object`</code>
14	PUT	<code>`/api/categories/{categoryId}`</code>	Updates a category belonging to the organiser's event	Organiser	<code>`{ categoryName, distanceKm, entryFee, maxParticipants }`</code>	<code>`200 OK` - updated category object`</code>
15	DELETE	<code>`/api/categories/{categoryId}`</code>	Removes a category from the organiser's event	Organiser	-	<code>`204 No Content`</code>

Event Enrolments						
16	POST	`/api/categories/{categoryId}/enrolments`	Enrols the logged-in participant into a category (generates a bib number)	Participant	`{}` (participant identified from token)	`201 Created` - `{ enrolmentId, bibNumber, status }`
17	GET	`/api/users/me/enrolments`	Lists all events/categories the logged-in participant has entered	Participant	-	`200 OK` - array of enrolments
18	GET	`/api/categories/{categoryId}/enrolments`	Lists all participants enrolled in a category, for the organiser managing that event	Organiser	-	`200 OK` - array of enrolments with participant details
19	DELETE	`/api/enrolments/{enrolmentId}`	Cancels the logged-in participant's own enrolment	Participant	-	`204 No Content`
Results						
20	POST	`/api/enrolments/{enrolmentId}/result`	Captures/records a finish result for an enrolment	Organiser	`{ finishTime, overallPosition, categoryPosition }`	`201 Created` - created result object
21	PUT	`/api/results/{resultId}`	Corrects a previously captured result	Organiser	`{ finishTime, overallPosition, categoryPosition }`	`200 OK` - updated result object
22	GET	`/api/events/{eventId}/results`	Lists all results for an event (public leaderboard)	Public	-	`200 OK` - array of results
23	GET	`/api/users/me/results`	Retrieves the logged-in participant's personal performance history	Participant	-	`200 OK` - array of the participant's past results

(Claude AI, 2026).

API Endpoint Plan drafted with assistance from Claude AI.

SQL Database Script

An SQL script that creates the full database schema for the RaceDay system.

```
CREATE DATABASE RaceDayDB;
```

```
USE RaceDayDB;
```

```
CREATE TABLE Roles(
```

```
  RoleID INT PRIMARY KEY,
```

```
  RoleName varchar(25) NOT NULL);
```

```
CREATE TABLE Users(
```

```
  UserID INT PRIMARY KEY, RoleID INT NOT NULL,
```

```
  FirstName varchar(30) NOT NULL, LastName varchar(30) NOT NULL,
```

```
  Email varchar(40) NOT NULL, PasswordHash varchar(100) NOT NULL,
```

```
  PhoneNumber varchar(30) NOT NULL, CreatedAt DATETIME NOT NULL,
```

```
CONSTRAINT FK_Users_Roles
```

```
FOREIGN KEY (RoleID) REFERENCES Roles(RoleID));
```

```
CREATE TABLE Enrollments(
```

```
  EnrolmentID INT PRIMARY KEY NOT NULL, ParticipantID INT NOT NULL,
```

CaategoryID INT NOT NULL, BibNumber varchar(10) NOT NULL,
EnrollmentDate DATETIME NOT NULL, Status varchar(15) NOT NULL,
CONSTRAINT FK_Enrollments_Users
FOREIGN KEY (ParticipantID) REFERENCES Users(UserID),
CONSTRAINT FK_Enrollments_Categories
FOREIGN KEY (CaategoryID) REFERENCES Categories(CaategoryID));

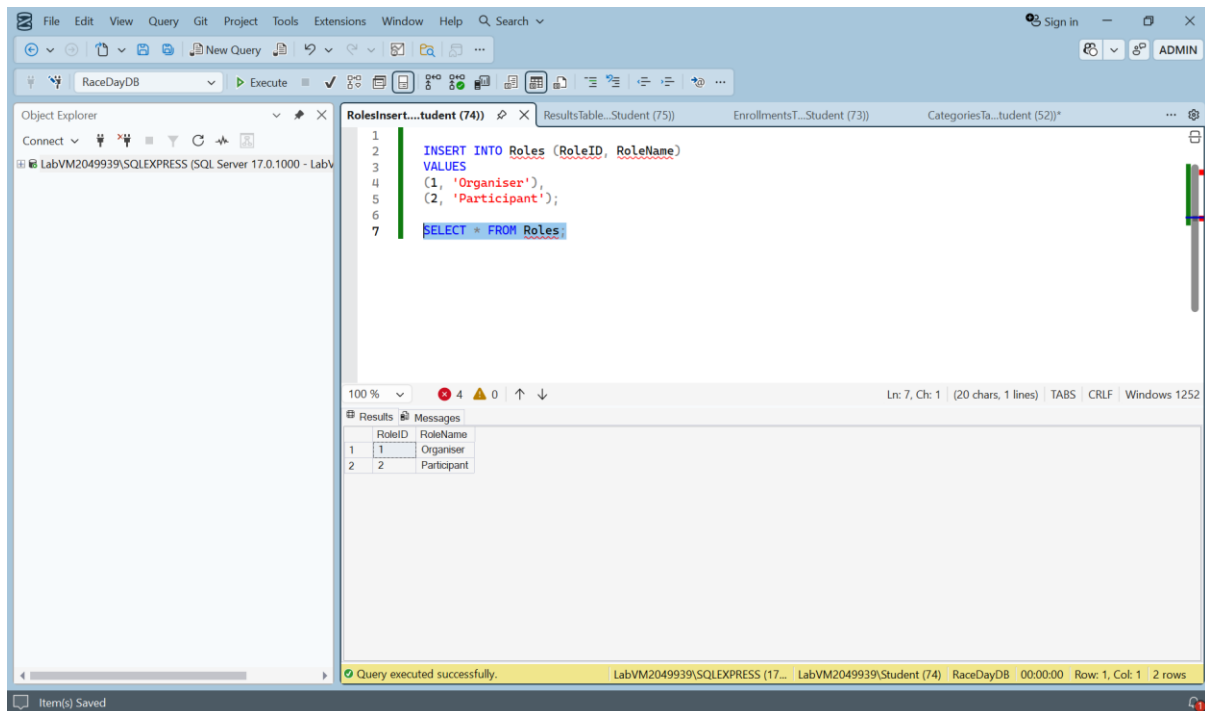
CREATE TABLE Categories(
CaategoryID INT PRIMARY KEY NOT NULL, EventID INT NOT NULL,
CategoryName varchar(100) NOT NULL, DistanceKM DECIMAL(5,2) NOT NULL,
EntryFee DECIMAL(8,2) NOT NULL, MaxParticipants INT NOT NULL,
CONSTRAINT FK_Categories_Events
FOREIGN KEY (EventID) REFERENCES Events(EventID));

CREATE TABLE Results(
ResultID INT PRIMARY KEY NOT NULL, EnrollmentID INT NOT NULL,
FinishTime TIME NOT NULL, OverallPosition INT NULL,
CategoryPosition INT NULL, RecordedAt DATETIME NOT NULL,
CONSTRAINT FK_Results_Enrollments
FOREIGN KEY (EnrollmentID) REFERENCES Enrollments(EnrollmentID));

CREATE TABLE Events(
EventID INT PRIMARY KEY NOT NULL, OrganiserID INT NOT NULL,
EventName varchar(100) NOT NULL, EventDate DATE NOT NULL,
Location varchar(100) NOT NULL, Description varchar(300) NOT NULL,
RouteInfo varchar(100) NULL, CreatedAt DATETIME NOT NULL,
CONSTRAINT FK_Events_Users
FOREIGN KEY (OrganiserID) REFERENCES Users(UserID));

INSERT INTO Roles (RoleID, RoleName)
VALUES
(1, 'Organiser'), (2, 'Participant');

SELECT * FROM Roles;



--Organisers--

INSERT INTO Users(UserID, RoleID, FirstName, LastName, Email, PasswordHash, PhoneNumber) VALUES (1,(SELECT RoleID FROM Roles WHERE RoleName='Organiser'), 'Sipho', 'Ndlovu', 'sipho.ndlovu@raceday.co.za', 'HASHED_PASSWORD1', '0711234567'), (2, (SELECT RoleID FROM Roles WHERE RoleName='Organiser'), 'Thandi', 'Mabasa', 'thandi.mabasa@raceday.co.za', 'HASHED_PASSWORD_2', '0845679832');

ALTER TABLE Users ADD CONSTRAINT DF_Users_CreatedAt DEFAULT GETDATE() FOR CreatedAt;

The screenshot shows the SQL Server Enterprise Manager interface. The query window displays the following SQL code:

```

1
2
3 --Organisers--
4 INSERT INTO Users(UserID, RoleID, FirstName, LastName, Email, PasswordHash, PhoneNumber)
5 VALUES
6 (1, (SELECT RoleID FROM Roles WHERE RoleName='Organiser'),
7 'Sipho', 'Ndlovu', 'sipho.ndlovu@raceday.co.za', 'HASHED_PASSWORD1', '0711234567'),
8 (2, (SELECT RoleID FROM Roles WHERE RoleName='Organiser'),
9 'Thandi', 'Mabasa', 'thandi.mabasa@raceday.co.za', 'HASHED_PASSWORD_2', '0845679832');
10
11 ALTER TABLE Users
12 ADD CONSTRAINT DF_Users_CreatedAt DEFAULT GETDATE() FOR CreatedAt;
13
14 SELECT * FROM Users;

```

The Results pane shows the output of the SELECT statement:

UserID	RoleID	FirstName	LastName	Email	PasswordHash	PhoneNumber	CreatedAt
1	1	Sipho	Ndlovu	sipho.ndlovu@raceday.co.za	HASHED_PASSWORD1	0711234567	2026-09-04 13:04:46.573
2	1	Thandi	Mabasa	thandi.mabasa@raceday.co.za	HASHED_PASSWORD_2	0845679832	2026-09-04 13:04:46.573

The status bar indicates: Query executed successfully. LabVM2049939\SQLEXPRESS (17... LabVM2049939\Student (60) RaceDayDB 00:00:00 Row: 1, Col: 1 2 rows

The screenshot shows the SQL Server Enterprise Manager interface. The query window displays the following SQL code:

```

1
2
3 --Participants--
4 INSERT INTO Users(UserID, RoleID, FirstName, LastName, Email, PasswordHash, PhoneNumber)
5 VALUES
6 (3, (SELECT RoleID FROM Roles WHERE RoleName='Participant'),
7 'Thabo', 'Mokoena', 'mokoenthabo@gmail.com', 'PASSWORD_HASH_4', '0679224112'),
8 (4, (SELECT RoleID FROM Roles WHERE RoleName='Participant'),
9 'Lindiwe', 'Khumalo', 'khumalolindi3@gmail.com', 'PASSWORD_HASH_5', '0636661648');
10
11 SELECT * FROM Users;

```

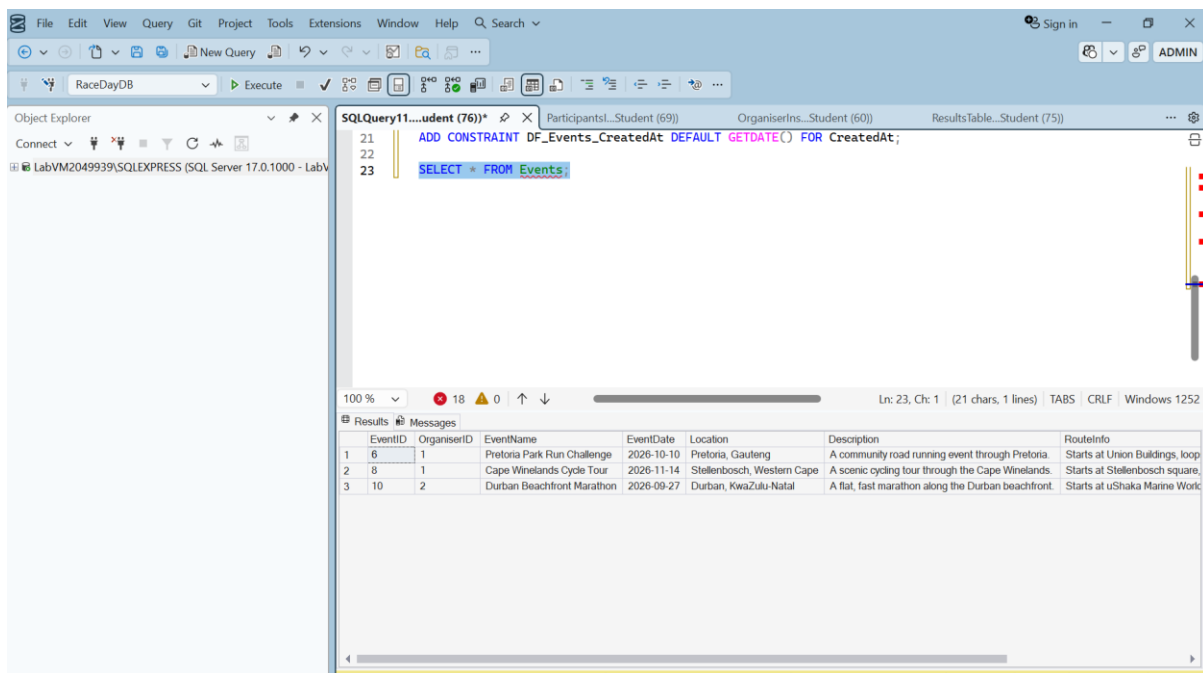
The Results pane shows the output of the SELECT statement:

UserID	RoleID	FirstName	LastName	Email	PasswordHash	PhoneNumber	CreatedAt
1	1	Sipho	Ndlovu	sipho.ndlovu@raceday.co.za	HASHED_PASSWORD1	0711234567	2026-09-04 13:04:46.573
2	1	Thandi	Mabasa	thandi.mabasa@raceday.co.za	HASHED_PASSWORD_2	0845679832	2026-09-04 13:04:46.573
3	2	Thabo	Mokoena	mokoenthabo@gmail.com	PASSWORD_HASH_4	0679224112	2026-09-04 13:15:24.257
4	2	Lindiwe	Khumalo	khumalolindi3@gmail.com	PASSWORD_HASH_5	0636661648	2026-09-04 13:15:24.257

The status bar indicates: Query executed successfully. LabVM2049939\SQLEXPRESS (17... LabVM2049939\Student (69) RaceDayDB 00:00:00 Row: 1, Col: 1 4 rows

INSERT INTO Events (EventID, OrganiserID, EventName, EventDate, Location, Description, RouteInfo)
VALUES
(6, (SELECT UserID FROM dbo.Users WHERE Email = 'sipho.ndlovu@raceday.co.za'),
'Pretoria Park Run Challenge', '2026-10-10', 'Pretoria, Gauteng',

'A community road running event through Pretoria.',
 'Starts at Union Buildings, loops through Arcadia Park.'),
 (8, (SELECT UserID FROM dbo.Users WHERE Email = 'sipho.ndlovu@raceday.co.za'),
 'Cape Winelands Cycle Tour', '2026-11-14', 'Stellenbosch, Western Cape',
 'A scenic cycling tour through the Cape Winelands.',
 'Starts at Stellenbosch square, loops through Franschhoek pass.'),
 (10, (SELECT UserID FROM dbo.Users WHERE Email =
 'thandi.mabasa@raceday.co.za'),
 'Durban Beachfront Marathon', '2026-09-27', 'Durban, KwaZulu-Natal',
 'A flat, fast marathon along the Durban beachfront.',
 'Starts at uShaka Marine World, runs along the promenade.');

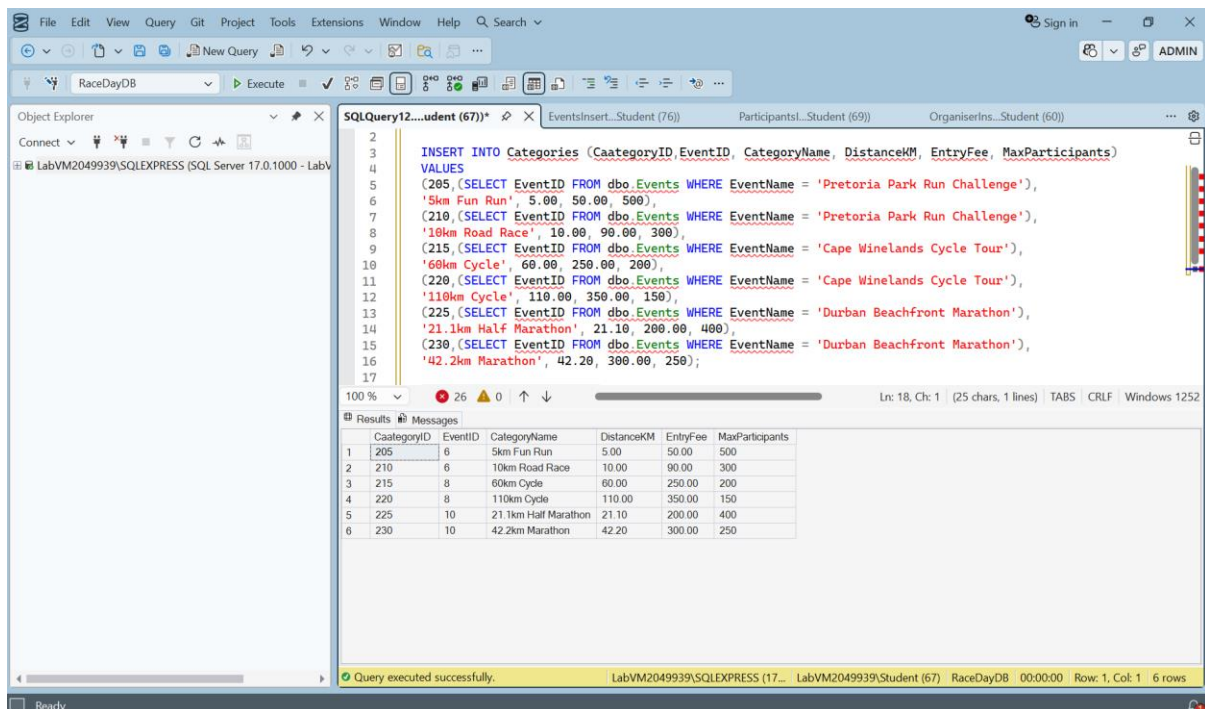


INSERT INTO Categories (CaategoryID,EventID, CategoryName, DistanceKM, EntryFee, MaxParticipants)
 VALUES
 (205,(SELECT EventID FROM dbo.Events WHERE EventName = 'Pretoria Park Run Challenge'),
 '5km Fun Run', 5.00, 50.00, 500),
 (210,(SELECT EventID FROM dbo.Events WHERE EventName = 'Pretoria Park Run Challenge'),
 '10km Road Race', 10.00, 90.00, 300),
 (215,(SELECT EventID FROM dbo.Events WHERE EventName = 'Cape Winelands Cycle Tour'),

```

'60km Cycle', 60.00, 250.00, 200),
(220,(SELECT EventID FROM dbo.Events WHERE EventName = 'Cape Winelands Cycle
Tour'),
'110km Cycle', 110.00, 350.00, 150),
(225,(SELECT EventID FROM dbo.Events WHERE EventName = 'Durban Beachfront
Marathon'),
'21.1km Half Marathon', 21.10, 200.00, 400),
(230,(SELECT EventID FROM dbo.Events WHERE EventName = 'Durban Beachfront
Marathon'),
'42.2km Marathon', 42.20, 300.00, 250);

```



```

INSERT INTO Enrollments (EnrollmentID, ParticipantID, CaategoryID, BibNumber,
Status)
VALUES
(1, (SELECT UserID FROM Users WHERE Email = 'mokoenaathabo@gmail.com'),
(SELECT CaategoryID FROM Categories WHERE CategoryName = '10km Road Race'),
'PTA1001', 'Confirmed'),
(2, (SELECT UserID FROM Users WHERE Email = 'khumalolindi3@gmail.com'),
(SELECT CaategoryID FROM Categories WHERE CategoryName = '21.1km Half
Marathon'), 'DBN2001', 'Confirmed')
(3, (SELECT UserID FROM Users WHERE Email = 'khumalolindi3@gmail.com'),
(SELECT CaategoryID FROM Categories WHERE CategoryName = '60km Cycle'),
'CPT3001', 'Confirmed');

```

SQLQuery13...udent (66)*

```

1  INSERT INTO Enrollments (EnrollmentID, ParticipantID, CaategoryID, BibNumber, Status)
2  VALUES
3  (1, (SELECT UserID FROM Users WHERE Email = 'mokoenthabo@gmail.com'),
4  (SELECT CaategoryID FROM Categories WHERE CategoryName = '10km Road Race'), 'PTA1001', 'Confirmed'),
5
6  (2, (SELECT UserID FROM Users WHERE Email = 'khumalolindi@gmail.com'),
7  (SELECT CaategoryID FROM Categories WHERE CategoryName = '21.1km Half Marathon'), 'DBN2001', 'Confirmed'),
8
9  (3, (SELECT UserID FROM Users WHERE Email = 'khumalolindi@gmail.com'),
10 (SELECT CaategoryID FROM Categories WHERE CategoryName = '60km Cycle'), 'CPT3001', 'Confirmed');
11
12 SELECT * FROM Enrollments;
13
14
15
16

```

EnrollmentID	ParticipantID	CaategoryID	BibNumber	EnrollmentDate	Status
1	3	210	PTA1001	2026-09-04 13:49:59.950	Confirmed
2	4	225	DBN2001	2026-09-04 13:49:59.950	Confirmed
3	4	215	CPT3001	2026-09-04 13:49:59.950	Confirmed

Query executed successfully. LabVM2049939\SQLEXPRESS (17... LabVM2049939\Student (66) RaceDayDB 00:00:00 Row: 1, Col: 1 3 rows

INSERT INTO Results (ResultID, EnrollmentID, FinishTime, OverallPosition, CategoryPosition)
VALUES
 (1, (SELECT EnrollmentID FROM dbo.Enrollments WHERE BibNumber = 'PTA1001'),
 '00:48:32', 12, 4);
SELECT * FROM Results;

SQLQuery15...udent (51)*

```

1  ALTER TABLE Results
2  ADD CONSTRAINT DF_Results_RecordedAt DEFAULT GETDATE() FOR RecordedAt;
3  INSERT INTO Results (ResultID, EnrollmentID, FinishTime, OverallPosition, CategoryPosition)
4  VALUES
5  (1, (SELECT EnrollmentID FROM dbo.Enrollments WHERE BibNumber = 'PTA1001'),
6  '00:48:32', 12, 4);
7
8  SELECT * FROM Results;

```

ResultID	EnrollmentID	FinishTime	OverallPosition	CategoryPosition	RecordedAt
1	1	00:48:32.0000000	12	4	2026-09-04 13:56:57.843

Query executed successfully. LabVM2049939\SQLEXPRESS (17... LabVM2049939\Student (51) RaceDayDB 00:00:00 Row: 1, Col: 1 1 rows